

BHARGAVI RAO BONDADA

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EDUCATION

M.S. Computer Science, AI, and HCI Focus · Boise State University, Boise, ID, *Expected Aug 2026*

B.Tech Computer Science · Vellore Institute of Technology, Chennai, India, 2022

TECHNICAL SKILLS

Languages and Frameworks: Python, Dart (Flutter), SQL (MySQL), JavaScript, Apache Spark, REST API Design, Object-Oriented Design, Firebase (Cloud Functions, Firestore, FCM, Auth)

AI and Agent Systems: AI Agent Development, LLM Integration, RAG, MCP Patterns, Prompt Engineering, Semantic Retrieval, BM25, Sentence Transformers, VLMs, OpenAI API, LiteLLM, SGLang, Ollama

Dev and Ops: Unit and Integration Testing, CI/CD Concepts, Deployment Workflows, AWS S3, EMR, Azure (AZ-900), Git, Figma, Jupyter Notebook

PROJECTS

Agentic Memory System, AI Agent with Persistent Memory | *Python · LLMs · RAG · MCP Patterns · VLMs · GPT-4o* (Co-developed with Anh and Amir)

- Designed and built a 1,000+ line AI agent system with persistent long-term memory: implemented multi-backend LLM orchestration (OpenAI, SGLang, Ollama, LiteLLM) via an abstract base class controller architecture, enabling plug-and-play backend swapping and clean separation of concerns across all agent components.
- Architected and implemented two RAG retrieval engines from scratch: a hybrid BM25 plus semantic search system and a pure embedding-based retriever (all-MiniLM-L6-v2), each configurable between top-k and similarity-threshold retrieval modes to address precision and recall tradeoffs across different query types.
- Applied MCP-style structured tool interfaces: enforced JSON-schema-validated LLM output contracts between agent components, eliminating unstructured output failures and hardening system reliability across all backend integrations.
- Integrated VLM captioning (GPT-4o, BLIP) to extend agent capabilities to multimodal image-aware queries; evaluated system on the LoComo benchmark across 5 question categories using F1, ROUGE-L, BLEU, BERT-F1, METEOR, and SBERT, with per-category analysis to isolate and address failure modes.

NudgeLab, Full-Stack Mobile Application | *Flutter · Firebase · Cloud Functions · Firestore · FCM · REST APIs · Figma* (github.com/BBhargaviRao/NudgeLab)

- Designed and shipped a full-stack Flutter/Firebase application: architected a Firestore NoSQL schema for real-time behavioral data persistence, built Firebase Cloud Functions handling server-side nudge scheduling, push notification dispatch via FCM, and trigger-based event logging across 10 child-parent participant pairs.
- Developed and deployed RESTful Cloud Function APIs with Firebase Auth for secure user management, structured data contracts for behavioral logging, and event-driven triggers, demonstrating API development and data lifecycle ownership in a live deployment.
- Owned full feature lifecycle from requirements gathering and Figma prototyping through iterative usability testing and redeployment: applied test-and-iterate feedback cycles, refining app behavior after each participant session to improve nudge engagement and interaction quality.

Wikipedia Graph Analysis Pipeline | *Apache Spark · Python · AWS S3 · EMR* (github.com/BBhargaviRao/wikipedia-spark-pipeline)

- Designed, deployed, and operated a two-part distributed backend pipeline on AWS EMR: Part 1 extracts bidirectional mutual links across 4 Wikipedia datasets; Part 2 computes connected components iteratively across millions of pages with S3 checkpointing for fault tolerance.
- Optimized Spark partitioning and transformation strategy to meet a sub-75-minute runtime SLA; built a synthetic unit test suite to validate pipeline correctness before production deployment, demonstrating testing discipline and observability ownership in a CI/CD-adjacent workflow.

Traffic Accident Severity Prediction | *Python · Scikit-learn · Pandas · Statistical Analysis*

- Built a scalable ML system from a 32-variable dataset: engineered features, trained a Random Forest classifier for multi-class severity prediction, and quantified actionable risk patterns, including nighttime incidents 35% more severe and combined adverse conditions driving 45% higher risk.
- Communicated model outputs through stakeholder-facing EDA visualizations (heatmaps, boxplots, trend charts), driving adoption of data-driven safety recommendations across non-technical audiences.

PUBLICATIONS · 3 PEER-REVIEWED

- Bondada et al. Soul Support: Designing Hopeful Wearables for Children's Emotional Wellness. ACM IDC '25, 2025.
- Bondada. Survey Trends and Taxonomy Distribution in Large Language Models. EngrXiv, 2024.
- Suhas P., Bondada B. et al. Object Detection using Different Point Feature Techniques. IJITEE, 2022.

CERTIFICATIONS AND RECOGNITION

- Microsoft Azure Fundamentals (AZ-900) · Microsoft
- McKinsey Forward Program · McKinsey.org (selective global analytics and leadership program)